

Elise Oltrogge

PH.D. CANDIDATE · PSYCHOLINGUISTICS

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Education

Goethe University Frankfurt

PH.D. PSYCHOLINGUISTICS

Frankfurt am Main, Germany

August 2023 - present

- Dissertation submitted on 07/09/2026
- First advisor: Prof. Dr. Sol Lago
- Second advisor: Prof. Dr. Shravan Vasishth

University of Potsdam

M.Sc. INTERNATIONAL EXPERIMENTAL AND CLINICAL LINGUISTICS

Potsdam, Germany

October 2020 - June 2023

- Nomination for the University of Potsdam Graduation Prize 2023

University of Potsdam

B.Sc. LINGUISTICS

Potsdam, Germany

October 2015 - June 2020

Skills

Data analysis

Frequentist and Bayesian statistics, generalized additive mixed models, simulation-based power analysis, data visualization

Programming Languages

R, Python, JavaScript, LISP, HTML, CSS, LaTeX, Markdown

Research Software

Experiment Builder, Data Viewer, PCIBex, OpenSesame, Montreal Forced Aligner, Praat, GitHub

Methodologies

Visual world eye-tracking, self-paced reading, timed production, reaction time experiments, mouse-tracking, cognitive modeling (ACT-R), computational modeling (Surprisal)

Languages

German (native), English (fluent), French (advanced), Spanish (basic)

Project Management

Event planning and organization, supervision of student research assistants

Professional Experience

2022–2023 Teaching Assistant for Prof. Dr. Shravan Vasishth in the Linguistics Department

Introduction to Statistical Data Analysis (M.Sc.)

Introduction to Statistics (B.Sc.)

- Opened each class with a review of the previous lecture
- Taught introductory R and RMarkdown in the first three lectures of each semester
- Held weekly office hours for statistical consultations
- Graded homework with written feedback
- Taught some lectures independently

2021–2023 **Student Assistant at the Vasishth Lab, University of Potsdam**

- Ran eye-tracking and electroencephalography (EEG) experiments
- Created stimuli, programmed and preprocessed web-based experiments using PClbex
- Recruited participants and managed studies on the SONA participant data base
- Assisted with setup of a new EEG device, organized test runs, and wrote protocols (PI: Dr. Pia Schoknecht)
- Led lab tours for incoming M.Sc. students, including live eye-tracking and EEG demonstrations
- Course support for the openHPI online course *Introduction to Bayesian Data Analysis* (2023): reviewed and tested materials, answered content and technical questions in the forum

2019–2021 **Student Assistant for Prof. Sol Lago (German Research Foundation Project AGREE)**

- Created experimental stimuli
- Programmed, preprocessed and ran eye-tracking and web-based experiments on Prolific
- Recruited participants and managed studies on the SONA participant data base
- Collected data from both native and second language speakers

2016–2019 **Archive Documentalist, N24 / WELT**

- Cataloged and documented video material from press agencies (APTN, Reuters) and daily news broadcasts using structured metadata and standardized descriptive language
- Ensured content was searchable and retrievable in the database for journalists and editors

Academic Training

STATISTICS & DATA ANALYSIS

- **Workshop on Bayesian Statistics and Poisson Regression**

Goethe University Frankfurt, 7–9 October 2025

Hands-on workshop on building, interpreting, and visualizing Bayesian mixed models using the *brms* package in R, with corpus and experimental language data.

- **Generalized Additive Mixed Models for Eye-Tracking Data**

Instructor: João Veríssimo. Goethe University Frankfurt, 6–7, 14 February 2025

Hands-on workshop on analyzing visual world eye-tracking data with GAMMs (mgcv). Covered fitting non-linear smooths, interactions between continuous and categorical predictors, random effects, and posterior simulation for temporal onset estimates.

- **Summer school on Statistical Methods for Linguistics and Psychology: Foundational methods in frequentist statistics.**

University of Potsdam, Germany. 9–13 September 2024

Week-long intensive course on the theory and application of frequentist statistics, with a focus on linear mixed models for linguistics and psychology.

- **Generalised Additive Models in R**

Physalia online course, 20–24 November 2023

Five-day course on using splines and GAMs to go beyond parametric modeling techniques such as GLMs. Covered model checking, selection, hierarchical GAMs, and posterior simulation.

- **Power analysis for mixed effects linear regression** (workshop)

Instructor: João Veríssimo. University of Lisbon, Online. 15 November 2023.

Workshop on simulation-based power analysis for studies using mixed effects linear regression.

- **Summer school on Statistical Methods for Linguistics and Psychology: Advanced Bayesian data analysis**

University of Potsdam, Germany. 12–16 September 2022.

Week-long intensive course on advanced Bayesian methods including hierarchical models and Stan, aimed at linguistics and psychology researchers.

- **Bayesian Data Analysis I** (6 ECTS, taken beyond required credits)

Instructor: Shravan Vasishth. University of Potsdam, Germany. Winter semester 2021/22. Grade: 1.3.

Graduate seminar covering the foundations of Bayesian data analysis, including prior specification, Markov chain Monte Carlo methods, and model comparison.

COMPUTATIONAL MODELING

- **Large Language Models in Psycholinguistics**

Goethe University Frankfurt, 14, 21 July 2025

Workshop on using large language models to calculate surprisal values for experimental materials in multiple languages, using Python in Google Colab.

- **Groningen Spring School on Cognitive Modeling: ACT-R, Nengo, PRIMs**

Groningen, the Netherlands, 8–12 April 2024

Five-day school covering computational cognitive modeling paradigms. Lectures and hands-on tutorials for developing process models of human cognition that can be tested against behavioral data.

EXPERIMENTAL METHODS & LAB TRAINING

- **Internship in Vasishth Lab**

Supervisor: Shravan Vasishth. University of Potsdam, Germany. October 2021–July 2022.

Hands-on lab internship involving participant recruitment, stimulus creation for EEG experiments, data collection (eye-tracking and EEG).

- **Experimental psychological training** (9 ECTS), M.Sc. Embodied Cognition

University of Potsdam, Germany. Winter semester 2017/18.

Designed, conducted, and wrote up a small experimental study using a grip force sensor.

RESEARCH SKILLS & PROFESSIONAL DEVELOPMENT

- **Research Data Management and Sharing**

The University of North Carolina at Chapel Hill & The University of Edinburgh. Coursera online course. September

2026.

Course on the research data lifecycle: data management plans, organization, documentation, storage, archiving, and sharing data through trustworthy repositories.

- **Getting Organized: Modern Knowledge Management for Researchers**

Goethe University Frankfurt, 23, 26 January 2026

Workshop on strategies and digital tools for organizing research literature, notes, and knowledge across projects.

- **Fast Forward. Project Management and Personal Productivity for Researchers**

Mind Matters, April–May 2024

Seven-week implementation program for an agile project and time management system tailored to research. Covered sprint planning, weekly reviews, and personal productivity routines.

- **Visualizing individual variability in language processing**

University of Potsdam, Germany. 13 January 2022.

Workshop on visualization techniques for individual-level data patterns in psycholinguistic experiments.

- **Introduction to Git and Github**

Google. Coursera online course. November 2021.

Course on version control with Git and collaborative workflows using GitHub.

- **Writing in the Sciences**

Stanford University. Coursera online course. July 2020.

Course on principles of clear scientific writing, the format of a scientific manuscript, peer review, and communicating science to general audiences.

Academic Service

Organizing committee 3–4 July 2025: *1st Workshop on Replication in the Language Sciences (WoReLa)*.
Goethe University, Frankfurt am Main

Journal reviewer *Journal of Memory and Language*

Abstract reviewer *Architectures and Mechanisms for Language Processing (AMLaP)*, *Computational Psycholinguistics Meeting (CPL)*, *Workshop on Replication in the Language Sciences (WoReLa)*

Publications

Oltrogge, E., Veríssimo, J., Patil, U., & Lago, S. (2025). Memory retrieval and prediction interact in sentence comprehension: an experimental evaluation of a cue-based retrieval model. *Journal of Memory and Language*, 144, 104651. <https://doi.org/10.1016/j.jml.2025.104651>

Oltrogge, E., & Patil, U (2025). Modeling Cue-Based Retrieval and Prediction — One Morpheme at a Time. In *Proceedings of the Annual Meeting of the Cognitive Science Society* (Vol. 47).

Lago, S., **Oltrogge, E.**, & Stone, K. (2025). Does productive agreement morphology increase sensitivity to agreement in a second language? *Glossa Psycholinguistics*, 4(1), 1–32. <https://doi.org/10.5070/g6011.25340>

- Lago, S., Schulz, P., Rinke, E., **Oltrogge, E.**, Dudschig, C., & Kaup, B. (2025). Insensitivity to truth-value in negated sentences: Does linear distance matter? *Experiments in Linguistic Meaning*, 3, 214–223. <https://doi.org/10.3765/elm.3.5797>
- Lago, S., Stone, K., **Oltrogge, E.**, & Veríssimo, J. (2023). Possessive processing in bilingual comprehension. *Language Learning*, 73(3), 904–941. <https://doi.org/10.1111/lang.12556>
- Stone, K., Veríssimo, J., Schad, D., **Oltrogge, E.**, Vasishth, S., & Lago, S. (2021). The interaction of grammatically distinct agreement dependencies in predictive processing. *Language, Cognition and Neuroscience*, 36(9), 1159–1179. <https://doi.org/10.1080/23273798.2021.1921816>

Presentations

INVITED TALKS

- Oltrogge, E. (2026). Are memory retrievals and predictions incremental at the morpheme-level? Talk at *Goethe University Frankfurt, Fiebach Lab*, January 14.
- Oltrogge, E. (2025). Prediction and retrieval in sentence comprehension: the role of word-internal features. Online talk at *University of Lisbon*, May 22.
- Oltrogge, E. (2025). Prediction and retrieval in sentence comprehension: the role of word-internal features. Online talk at *University of Hull*, April 30.
- Oltrogge, E. (2024). Prediction as retrieval: An extension of a cue-based retrieval model of German possessive pronouns. Talk at *University of Potsdam*, June 13.
- Oltrogge, E. (2022). Cue interaction in predictive processing of German possessive pronouns. Online talk at *University of Lisbon*, May 31.

CONFERENCE PRESENTATIONS

- Oltrogge, E.**, Lago, S., Veríssimo, J. & Patil, U. (2026). Predictions beyond the word: Evidence for morpheme-level incrementality from cognitive modeling and eye-tracking. Talk at *AMLaP*, Saarbrücken, Germany, September 2–4.
- Gomes, J., Martinez Bruera, M., Parrish, K. & **Oltrogge, E.** (2026). The effect of coarticulation on L2 spoken word recognition. Poster at *AMLaP*, Saarbrücken, Germany, September 2–4.
- Gomes, J., Martinez Bruera, M., Parrish, K. & **Oltrogge, E.** (2026). Coarticulatory cues in L2 spoken word recognition. Poster at *WoProc*, Lisbon, Portugal, July 6–8.
- Gomes, J., Martinez Bruera, M., Parrish, K. & **Oltrogge, E.** (2026). Can L2 speakers use coarticulatory cues in spoken word recognition? Poster at *EuroSLA*, Lisbon, Portugal, June 24–27.
- Gomes, J., Martinez Bruera, M., Parrish, K. & **Oltrogge, E.** (2026). The effect of coarticulation on L2 spoken word recognition. Poster at *HSP*, Cambridge, Massachusetts, March 26–28.
- Oltrogge, E.**, Lago, S., Veríssimo, J. & Patil, U. (2026). How incremental are predictions? Insights from cognitive modeling and eye-tracking. Talk at *HSP*, Cambridge, Massachusetts, March 26–28.

- Oltrogge, E.** & Patil, U. (2026). Modeling cue-based retrieval and prediction based on morpheme-level cues. Poster at *Computational Psycholinguistics Meeting 2025*, Utrecht, Netherlands, December 18–19.
- Oltrogge, E.**, Lee, E.-K., & Lago, S. (2025). Can planned words trigger interference during real-time sentence production? Poster at *AMLaP*, Prague, the Czech Republic, September 4–6.
- Lee, E.-K., **Oltrogge, E.** & Lago, S. (2025). Can planned words trigger interference during real-time sentence production? Poster at *HSP*, College Park, Maryland, March 27–29.
- Oltrogge, E.**, Veríssimo, J., Patil, U., & Lago, S. (2025). Unifying memory retrieval and prediction in sentence comprehension: a cue-based computational model, Poster at *HSP*, College Park, Maryland, March 27–29.
- Oltrogge, E.**, Veríssimo, J., Patil, U., & Lago, S. (2024). Memory retrieval influences the time course of predictions during pronoun processing. Poster at *AMLaP*, Edinburgh, Scotland, September 5–7.
- Lago, S. Schulz, P., Rinke, E., **Oltrogge, E.**, Dudschig, C., & Kaup, B. (2024). Insensitivity to truth-value in negated sentences: does linear distance matter? Online talk at *Experiments in Linguistic Meaning (ELM3)*, June 12.
- Oltrogge, E.**, Veríssimo, J., Vasishth, S., & Lago, S. (2023). Do comprehenders use cognitive control to resolve morphosyntactic conflicts during prediction? Poster at *AMLaP*, San Sebastian, Spain, August 31–September 2.
- Lago, S., Stone, K., & **Oltrogge, E.** (2021). Illusions in bilingual gender predictions. Talk at *LISP 2021: Linguistic Illusions in Sentence Processing*, Constance, Germany, September 13–14.
- Oltrogge, E.**, & Lago, S. (2021). Proficiency and between-language congruency interact in L2 gender acquisition. Poster at *AMLaP*, Paris, France, September 2–4.
- Lago, S., **Oltrogge, E.**, & Stone, K. (2020). Are native and non-native speakers differentially sensitive to agreement attraction? Poster at *AMLaP*, Potsdam, Germany, September 3–5.
- Stone, K., **Oltrogge, E.**, Vasishth, S. & Lago, S. (2020). The real-time application of grammatical constraints to prediction: Timecourse evidence from eye tracking. Talk at *CUNY*, Amherst, Massachusetts, March 19–21.

References

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